Using Normalizing Flows for Weighting Estimators with Continuous Treatment

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Abstract

1 Introduction

In observational studies, political scientists care about an unbiased estimator for average treatment effect (ATE) or average treatment effect on the treated (ATT). One way to generate an unbiased and also semiparametrically efficient estimator is by using augmented inverse propensity weighting(AIPW) estimator, known to be doubly robust. However, there are two caveats here. First, as continuous treatment is common in political science studies, simply dichotomizing treatments to fit into AIPW with a binary treatment framework may lose information on the treatment effect heterogeneity. Second, although researchers can use logit or probit to estimate the propensity score with binary treatment easily, an accurate estimation of the propensity score under continuous treatment is still a challenge. In this paper, we propose to use the normalizing flows (NF), which produce efficient and exact propensity score estimation with continuous treatment.

2 Related Literature

2.1 Continuous Treatment and Generalized Propensity Score

A large literature studies the inference for various treatment effects with binary or categorical treatments, either with randomized experiment or observational studies. In political science, such examples include the Get Out the Vote (GOTV) campaign on voter turnout (Gerber and Green, 2000), the effect of product differentiation in economic markets on firm-level lobbying in political markets (Kim, 2017), and the effect of patron-client network on economic development in China (Jiang, 2018). However, many empirical settings could also involve continuous treatment, such as the campaign spending, the intensity of a policy implementation, as well as the media exposure. In such settings, causal effects are naturally summarized by an average dose-response function, or the average structural function (Colangelo and Lee,

2026).

$$m(t) = \mathbb{E}\left[Y(t)\right],$$

rather than by a single average treatment effect. Under consistency, overlap, and conditional ignorability assumption, this curve can be identified as

$$m(t) = \mathbb{E}\left[\mu(X,t)\right], \qquad \mu(x,d) = \mathbb{E}\left[Y \mid X = x, T = t\right].$$

A strand of literature focuses on the treatment density estimation. Hirano and Imbens (2004) provide the foundational propensity-score approach for continuous treatment: the generalized propensity score (GPS). Defined as

$$g(t \mid x) = f_{T|X}(t \mid x),$$

GPS is the conditional density of the treatment given the covariates. They have shown that the GPS plays an analogous balancing role to the binary propensity score. Although their framework provides a way to estimate does-response functions by adjusting for the GPS, it often relies on parametric modeling of the treatment density. Imai and van Dyk (2004) extend the idea of GPS through the more general framework of a propensity function for treatment regimes beyond the binary case. Their framework clarifies that causal adjustment with non-binary treatment requires modeling the treatment assignment process as a conditional distribution, not merely as a treatment probability.

Another line of research focuses on the covariate balance. Fong et al. (2018) develop the covariate balancing GPS for continuous treatments. Their method estimates weights that reduce association between treatment and covariates, extending the covariate balancing propensity score logic to continuous exposures. This approach highlights the limitation of standard GPS estimation: a well-fitting parametric treatment model does not necessarily

imply good covariate balance.

Taken together, this literature establishes that the GPS is a central project in the casual inference with continuous treatment. However, as the GPS is a conditional density, its estimation can be statistically challenging and it could be vulnerable to model misspecification. Such limitation motivates a move from propensity-score adjustment alone to semiparametric methods that combine the treatment-density model with an outcome regression model.

2.2 Semiparametric Theory and Doubly Robust Estimators

Semiparameric theory provides a general framework for constructing estimators that combine multiple nuisance functions while preserving valid inference for a lower-dimensional target parameter. In causal inference, this framework is especially useful because most causal estimands depend on nuisance components such as the outcome regression, the treatment assignment mechanism, and the covariate distribution. Efficient influence functions play a central role here. They characterize the first-order sensitivity of the target parameter to perturbations of the observed-data distribution and often lead directly to bias-corrected or augmented estimators.

For binary treatment, this logic yields the AIPW estimator with the double robustness property, which means.

As mentioned by Kennedy et al. (2017), semiparametric theory is not directly available if we only assume mild smoothness conditions on $\theta(d)$ (e.g., differentiability) since without parametric assumptions $\theta(d)$ is not pathwise differentiable, and root n consistent estimators do not exist. By defining a parameter that represents the average outcome under an intervention that randomly assigns treatment based on the density of the continuous treatment, Kennedy et al. (2017) construct the efficient influence function for this stochastic-intervention parameter, which contains a pseudo-outcome term that attains the double robustness.

3 Problem Setup

4 Normalizing Flows

NF are generative models which produce tractable distributions where both sampling and density evaluation can be efficient and exact, using invertible functions. In other words, it is a technique that maps a random variable with a complex probability density function to the simpler density function.

A typical NF procedure starts with a simple base distribution such as Gaussian and then attempts to transform that function into a more complex distribution by applying a finite number of invertible functions[1]. The process utilized the *change of variables formula*, which guarantees mapping of one probability density function for a random variable Y with another probability density function of random variable Z:

$$\begin{aligned} p_{\mathbf{Y}}(\mathbf{y}) &= p_{\mathbf{Z}}(\mathbf{f}(\mathbf{y}))|\det \mathrm{D}\,\mathbf{f}(\mathbf{y})| \\ &= p_{\mathbf{Z}}(\mathbf{f}(\mathbf{y}))|\det \mathrm{D}\,\mathbf{g}(\mathbf{f}(\mathbf{y}))|^{-1}, \end{aligned}$$

where $\mathbf{g}$ is an invertible function, $\mathbf{f} = \mathbf{g}^{-1}$, and $\mathrm{D}\,\mathbf{f}(\mathbf{y}) = \frac{\partial \mathbf{f}}{\partial \mathbf{y}}$ is the Jacobian of $\mathbf{f}$. Figure 1 illustrates this relationship. Since $\mathbf{f}$ moves in the direction of a simple base distribution, it is called *normalizing flow*.

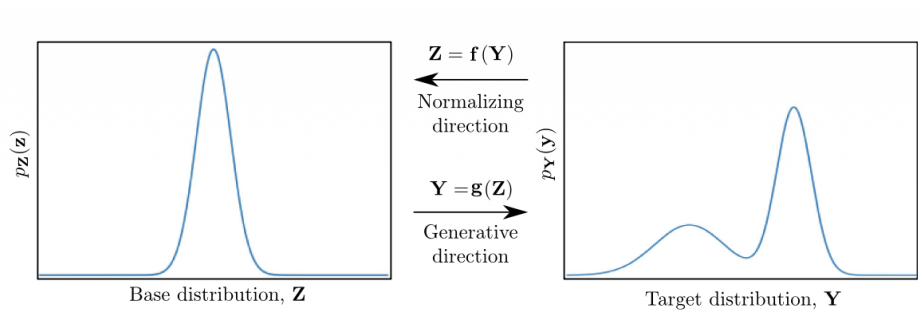

Figure 1: Graphical Illustration of Normalizing Flows. from Kobyzev et al. (2021)

[1]The existence of this process is always guaranteed, and discussed in Chen and Gopinath (2000).

As a result of this process, we can link the complex distribution (such as a joint distribution of variables we observe) with simple base distribution such as Gaussian. Since this is an invertible process, instead of dealing with the complex distribution itself, we can work with the simple base distribution such as sampling and density estimation, and then easily bring the result to the complex one.

NF is a useful tool in density estimation compare to other frequently used density estimation method. Histogram is a simple exploratory method. However, it relies heavily on the size of bins and thus often performs poorly on complex distributions. Unlike the histogram, KDE technique produces a smooth estimate of the density, using all sample points' locations. The main idea behind KDE has been proposed long ago (Rosenblatt, 1956), but the recent development of computing technology makes it a popular option today. KDE, however, requires researchers to decide which kernel to be used and the size of the bandwidth, and both influence the estimation result. Since NF usually relies on neural network, the computational cost may be high for complex design, it is more capable of dealing with complex density.

5 Using NF for Propensity Score Estimation

One straight forward application of density estimation with NF is propensity score estimation. Statistical estimators employing propensity score relies heavily on the accuracy of the propensity score.

Although there are criticism on using machine learning algorithm in propensity score estimation, it becomes more popular (Goller et al., 2020). Lechner and Okasa (2024) utillized random forest estimation of propensity score. In this paper, we propose to use NF as an alternative for propensity score estimation and leverage the estimated propensity score into the AIPW estimator in observational studies.

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